

Feiya Ou

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OBJECTIVE

I seek to uncover fundamental mechanisms of transcription and aspire to become a leader at the interface of genetics and immunology.

EDUCATION

Washington University in St. Louis (WashU)

St. Louis, MO, USA

Ph.D. in Immunology

Sep 2019–Present

Advisor: Dr. Kenneth M. Murphy

Thesis: *Molecular Circuitries in Dendritic Cell Development*

Xiamen University

Xiamen, Fujian, China

B.S. in Biology

Sep 2015–Jun 2019

AWARDS & HONORS

- **FASEB Journal Early Career Researcher Award** (2025): Sole recipient at the conference; awarded for presenting rigorous, impactful research findings that advance the field of science at a FASEB Science Research Conference
- **Lucille P. Markey Special Emphasis Pathway in Human Pathobiology** (2020 – 2022): a selective program designed to inspire fundamental research related to human disease, including coursework in pathobiology and clinical shadowing experiences
- **Talented Undergraduate Training Program of the Beutler Institute** (*ranked first in 20 selected students*; 2018 – 2019): a highly selective program initiated by Nobel Laureate Dr. Bruce Beutler, which included a semester of advanced coursework in Genetics and Immunology at Xiamen University, followed by a year of scientific training at UT Southwestern Medical Center
- **Pilot Program for Cultivating Top-notch Students in Basic Disciplines** (2016 – 2019)

PUBLICATIONS (see also [Google Scholar](#))

Under Review

1. **Ou, F.**, Liu, T.-T., Du, S., Chen, J., Koch, A. R., Kraft, M., Murphy, T. L., & Murphy, K. M. (2025). ID2 secures cDC1 specification by antagonizing E proteins at a pleiotropic Zeb2 enhancer. In Res. Sq. <https://doi.org/10.21203/rs.3.rs-7455813/v1> (under revision at *Nature Immunology*)
2. Du, S., **Ou, F.**, Drieu, A., Cheng, Y., Storck, S. E., Rustenhoven, J., Mamuladze, T., Bhattarai, B., Brioschi, S., Nguyen, K., Cao, J., Rodrigues, P. F., Smirnov, I., DeNardo, D., Ginhoux, F., Cella, M., Colonna, M., & Kipnis, J. (2024). Brain-engrafted monocyte-derived macrophages from blood and skull-bone marrow exhibit distinct identities from microglia. In bioRxiv.org <https://doi.org/10.1101/2024.08.08.606900> (under review at *Neuron*)

Published

1. Chen, J., Liu, T.-T., **Ou, F.**, Ohara, R. A., Jo, S., Postoak, J. L., Egawa, T., Day, R. B., Murphy, T. L., Murphy, K. M., & Kim, S. (2025). C/EBP α activates Irf8 expression in myeloid progenitors at the +56-

- kb enhancer to initiate cDC1 development. *Science Immunology*, 10(107), eadt5899. <https://doi.org/10.1126/sciimmunol.adt5899>
2. Jo, S., Ohara, R. A., Theisen, D. J., Kim, S., Liu, T., Bullock, C. B., He, M., **Ou, F.**, Chen, J., Piersma, S. J., Postoak, J. L., Yokoyama, W. M., Diamond, M. S., Murphy, T. L., & Murphy, K. M. (2025). Shared pathway of WDFY4-dependent cross-presentation of immune complexes by cDC1 and cDC2. *The Journal of Experimental Medicine*, 222(4). <https://doi.org/10.1084/jem.20240955>
3. **Ou, F.**, & Murphy, K. M. (2025). “What’s in a name?” Clarifying the identity of ROR γ t⁺ antigen-presenting cells. *The Journal of Experimental Medicine*, 222(8), e20250760. <https://doi.org/10.1084/jem.20250760>
4. Postoak, J. L., Koch, A., Shan, S., Ohara, R. A., Jo, S., Chen, J., Desai, P., **Ou, F.**, Kraft, M., Diamond, M. S., Kim, S., Murphy, T. L., & Murphy, K. M. (2025). WDFY4-dependent cross-presentation proceeds via a vacuolar antigen-processing route. *Proceedings of the National Academy of Sciences of the United States of America*, 122(50), e2519922122. <https://doi.org/10.1073/pnas.2519922122>
5. Kim, S., Chen, J., **Ou, F.**, Liu, T.-T., Jo, S., Gillanders, W. E., Murphy, T. L., & Murphy, K. M. (2024). Transcription factor C/EBP α is required for the development of Ly6Chi monocytes but not Ly6Clo monocytes. *Proceedings of the National Academy of Sciences of the United States of America*, 121(15), e2315659121. <https://doi.org/10.1073/pnas.2315659121>
6. Kim, S., Liu, T.-T., **Ou, F.**, Murphy, T. L., & Murphy, K. M. (2024). Anatomy of a superenhancer. *Advances in Immunology*, 163, 51–96. <https://doi.org/10.1016/bs.ai.2024.08.001>
7. Murphy, K. M., & **Ou, F.** (2024). Weak enhancer allows autoactivation of Irf8 to control cDC1 versus cDC2 lineage commitment. *Nature Immunology*, 25(11), 1997–1998. <https://doi.org/10.1038/s41590-024-01977-9>
8. **Ou, F.**, Liu, T.-T., Desai, P., Ferris, S. T., Kim, S., Shen, H., Ohara, R. A., Jo, S., Chen, J., Postoak, J. L., Du, S., Diamond, M. S., Murphy, T. L., & Murphy, K. M. (2024). Optimization of the Irf8 +32-kb enhancer disrupts dendritic cell lineage segregation. *Nature Immunology*, 25(11), 2043–2056. <https://doi.org/10.1038/s41590-024-01976-w>
9. Ferris, S. T., Liu, T., Chen, J., Ohara, R. A., **Ou, F.**, Wu, R., Kim, S., Murphy, T. L., & Murphy, K. M. (2023). WDFY4 deficiency in NOD mice ameliorates autoimmune diabetes and insulinitis. *Proceedings of the National Academy of Sciences of the United States of America*, 120(13), e2219956120. <https://doi.org/10.1073/pnas.2219956120>
10. Kim, S., Chen, J., Jo, S., **Ou, F.**, Ferris, S. T., Liu, T.-T., Ohara, R. A., Anderson, D. A., Wu, R., Chen, M. Y., Gillanders, W. E., Gillanders, W. E., Murphy, T. L., & Murphy, K. M. (2023). IL-6 selectively suppresses cDC1 specification via C/EBP β . *The Journal of Experimental Medicine*, 220(10), e20221757. <https://doi.org/10.1084/jem.20221757>
11. Liu, T.-T., **Ou, F.**, Belk, J. A., Bagadia, P., Anderson, D. A., 3rd, Durai, V., Yao, W., Satpathy, A. T., Murphy, T. L., & Murphy, K. M. (2023). Cis interactions in the Irf8 locus regulate stage-dependent enhancer activation. *Genes & Development*, 37(7–8), 291–302. <https://doi.org/10.1101/gad.350339.122>
12. **Ou, F.**, Ferris, S. T., Kim, S., Wu, R., Anderson, D. A., 3rd, Liu, T.-T., Jo, S., Chen, M. Y., Gillanders, W. E., Murphy, T. L., & Murphy, K. M. (2023). Enhanced in vitro type 1 conventional dendritic cell generation via the recruitment of hematopoietic stem cells and early progenitors by Kit ligand. *European Journal of Immunology*, 53(9), e2250201. <https://doi.org/10.1002/eji.202250201>
13. Anderson, D. A., **Ou, F.**, Kim, S., Murphy, T. L., & Murphy, K. M. (2022). Transition from cMyc to L-Myc during dendritic cell development coordinated by rising levels of IRF8. *The Journal of Experimental Medicine*, 219(2). <https://doi.org/10.1084/jem.20211483>
14. Ferris, S. T., Ohara, R. A., **Ou, F.**, Wu, R., Huang, X., Kim, S., Chen, J., Liu, T.-T., Schreiber, R. D., Murphy, T. L., & Murphy, K. M. (2022). cDC1 vaccines drive tumor rejection by direct presentation independently of host cDC1. *Cancer Immunology Research*, 10(8), 920–931. <https://doi.org/10.1158/2326-6066.CIR-21-0865>

15. Liu, T.-T., Kim, S., Desai, P., Kim, D.-H., Huang, X., Ferris, S. T., Wu, R., **Ou, F.**, Egawa, T., Van Dyken, S. J., Diamond, M. S., Johnson, P. F., Kubo, M., Murphy, T. L., & Murphy, K. M. (2022). Ablation of cDC2 development by triple mutations within the Zeb2 enhancer. *Nature*, 607(7917), 142–148. <https://doi.org/10.1038/s41586-022-04866-z>
16. Wu, R., Ohara, R. A., Jo, S., Liu, T.-T., Ferris, S. T., **Ou, F.**, Kim, S., Theisen, D. J., Anderson, D. A., 3rd, Wong, B. W., Gershon, T., Schreiber, R. D., Murphy, T. L., & Murphy, K. M. (2022). Mechanisms of CD40-dependent cDC1 licensing beyond costimulation. *Nature Immunology*, 23(11), 1536–1550. <https://doi.org/10.1038/s41590-022-01324-w>
17. Misawa, T., SoRelle, J. A., Choi, J. H., Yue, T., Wang, K.-W., McAlpine, W., Wang, J., Liu, A., Tabeta, K., Turer, E. E., Evers, B., Nair-Gill, E., Poddar, S., Su, L., **Ou, F.**, Yu, L., Russell, J., Ludwig, S., Zhan, X., ... Beutler, B. (2020). Mutual inhibition between Prkd2 and Bcl6 controls T follicular helper cell differentiation. *Science Immunology*, 5(43), eaaz0085. <https://doi.org/10.1126/sciimmunol.aaz0085>
18. Turer, E. E., San Miguel, M., Wang, K.-W., McAlpine, W., **Ou, F.**, Li, X., Tang, M., Zang, Z., Wang, J., Hayse, B., Evers, B., Zhan, X., Russell, J., & Beutler, B. (2018). A viable hypomorphic Arnt2 mutation causes hyperphagic obesity, diabetes and hepatic steatosis. *Disease Models & Mechanisms*, 11(12), dmm035451. <https://doi.org/10.1242/dmm.035451>

PRESENTATIONS

Oral Presentations

- 2025 WashU Immunology Program Work-in-Progress (*opening seminar*)
- 2025 FASEB Mechanisms of Immune Cell Development and Function (*FASEB Journal Early Career Researcher award*)
- 2025 WashU Genetics Work-in-Progress
- 2024 WashU Immunology Program Annual Retreat
- 2024 WashU Immunology Program Work-in-Progress (*recruitment seminar*)
- 2022 WashU Markey Pathway Annual Retreat

Poster Presentations

- 2025 Cold Spring Harbor Laboratory (CSHL) Mechanisms of Eukaryotic Transcription
- 2024 4D Nucleome (4DN) Annual Meeting
- 2024 CSHL Gene Expression & Signaling in the Immune System
- 2022 CSHL Gene Expression & Signaling in the Immune System

RESEARCH SKILLS

- **Mouse genetics:** CRISPR/Cas9 genome editing, ES cell-based homologous recombination, breeding, genotyping, and colony management
- **Molecular cloning**
- **Flow cytometry and FACS sorting**
- **Cell culture:** established cell lines and primary bone marrow-derived immune cell cultures
- **Genomics assays and analyses:** CUT&RUN, bulk RNA-seq, and single-cell RNA-seq
- **Biochemical assays:** gel shift assay (EMSA) and Southern blot

TEACHING EXPERIENCE

Teaching Assistant, *BIO 5053 Immunobiology I* (graduate level), WashU, Fall 2020

Course Director: Dr. Takeshi Egawa

Enrollment: 29 students

Led weekly review sections; drafted and graded assignments and three exams

MENTORSHIP EXPERIENCE

- **Dr. Magdalena Kraft:** Dr. Kraft is a postdoctoral researcher who joined the Murphy lab in 2024. Since her arrival, I have been actively involved in her training.
- **Ms. Alyssa R. Koch:** Alyssa joined the Murphy lab after completing her rotation as a student in the Immunology program in 2025. I began mentoring her during her rotation and am continuing to provide training and guidance.
- **Ms. Wenxuan Cheng:** Wenxuan joined the Murphy lab in 2025 after completing her rotation as a student in the Immunology program in 2024. I began mentoring her during her rotation and am continuing to provide training and guidance.
- **Mr. Haolin (Charles) Shen:** Charles was an undergraduate student who majored in Biology. I mentored him throughout his research in the Murphy lab in 2022 -2023. He received his BA in 2023 and is now a Ph.D. student in the Biomedical Sciences Program at the University of California, San Francisco, in Dr. Jason G. Cyster's lab.

LEADERSHIP & SERVICE

- Participated in multiple mock qualifying exams to help fellow PhD students across programs prepare for their prelims (2021–present)
- Founded and led a regular basketball group for graduate students and postdocs at the WashU Medical Campus (2022–present)

PEER REVIEW & EDITORIAL SERVICE

- Topic coordinator: *Frontiers in Immunology*
- *Ad hoc* reviewer: *The Journal of Experimental Medicine*; *Nature Communications*